

Theory

① What is inferential statistics?

Ans:- It is type of statistics, it is used to make prediction and draw conclusions about your data.

* Its key concepts:-

* hypothesis testing

* confidence interval

* correlation

② What is hypothesis testing and its components?

Ans:- Hypothesis testing is stat. method to make decision about population parameters based on sample data.

* Its components:-

* Null hypothesis (H_0)

* Alternate hypothesis (H_1 or H_a)

③ Explain confidence interval and critical value?

Ans:- confidence interval gives you range of values that likely to lie population parameters based on sample data.

* critical value is threshold value which tell you if your test statistics value above this value then your H_0 is rejected otherwise we fail to reject H_0 .

④ Define P value?

Ans:- P value stand for probability, which is find using various test and it is compare with significance value to determine our H_0 is rejected or Fail to reject.

⑤ Differentiate between type I and II error?

Ans:- Type I error:-

* This is occur when we know our H_0 is true although we reject H_0 .

* Type II error:-

* This is occur when we know our H_1 is true although we reject H_1 .

⑥ Brief descriptions of z test, t test, chi-square test and ANOVA test?

Ans:- z test check if sample mean is significantly different from known population mean.

* t test compare mean of two groups to see if they're significantly different.

* chi-square test check a significant association between two categorical variables.

* ANOVA test compare three or more groups to see if there's significant difference.

⑦ What is covariance?

Ans:- covariance measure how much two variables change together. If they tend to increase or decrease, covariance is positive. If one increase and other decrease then covariance is negative.

⑧ What is correlation?

Ans:- correlation measure strength and direction of the relationship between two variables. its range value range is -1 to 1:-

- * -1 means perfect negative relationship
- * 1 means perfect positive relationship
- * 0 means no relationship